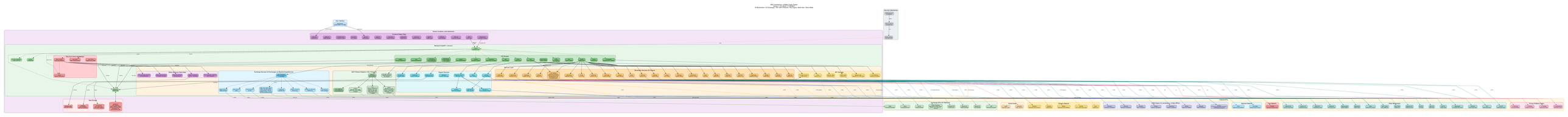

ABCT Architecture - A Better Crypto Tracker

Version 1.13.0 (Build 1771554807)

34 Blockchains | 42 Exchanges | 100+ DeFi Protocols | P&L Engine | Multi-User | Demo Mode

User Interface
Web Browser (localhost:8000 / Port 8080)
HTTP/HTTPS GET /
Authentication
Docker Container: abct-dashboard
Frontend (Static Files)
index.html Dashboard UI
wallets.html Wallet Manager
transactions.html Transaction History
nft-wall.html NFT Gallery
login.html Authentication
apis.html API Manager
security.html Security Settings
services.html Health Status
backup.html Backup & Restore
logs.html System Logs
cache.html Cache Manager
styles.css Theming (5 themes)
app.js Client Logic
transactions.js Transaction Analytics
REST API JSON
Transaction API
Backend (FastAPI + Uvicorn)
main.py Application Entry
scans
nft_image_database.py Image Cache DB
config.py Configuration
Security & Demo Middleware
demo_mode.py Demo User Detection
rate_limit.py Rate Limiting (slowapi)
size_limit.py Request Size (10MB)
auth.py Session Authentication
session check
API Routers
/settings API Key Management
/auth Login/Logout/Sessions
/exchanges 42 Exchange APIs + ExchangeRegistry
/defi/protocols Protocol Detection + P&L Analytics
/security SSL/Certificate Mgmt
/portfolio Summary & History
/transactions Multi-Chain Txn History
/defi DeFi Positions
/prices Crypto Prices
/nfts NFT Collections
/custom-tokens Token Tracking
/pnl Cost Basis P&L Reporting
/logs System Logging
/wallets Wallet CRUD
/backup Export/Import
/nft-scheduler Background NFT Updates
demo user
API keys
sessions
SSL config
ExchangeRegistry
Demo Services (Fake Data)
demo_exchange_service.py Exchange Balances
demo_wallet_service.py 81 Tokens, 26 Chains
demo_defi_service.py Staking Positions
demo_nft_service.py 76 NFTs, 7 Collections
demo_transaction_service.py 1,000 Transactions
Exchange Services (42 Exchanges via BaseExchangeService)
base_exchange.py BaseExchangeService + 5 Auth Mixins
+30 more
bitget_service.py Bitget Exchange
gate_service.py Gate.io
kucoin_service.py KuCoin
bybit, mexc, kraken, gemini bitfinex, htx, bingx, phemex woox, poloniex, cryptocom lbitstamp, bitmart + 22 more (35 new exchange services)
coinbase_service.py Coinbase CDP
binance_service.py Binance.com
binance_us_service.py Binance.US
okx_service.py OKX Exchange
DeFi Protocol Adapters (100+ Protocols)
registry.py ProtocolRegistry (auto-discovery)
base_adapter.py ProtocolAdapter ABC 5 DetectionMethods
cardano/ Minswap, Liqwid, Indigo Strike, Surf (UTXO_SCAN) 5 adapters
evm/ Aave v3, Compound v3 Uniswap v3 LP, Curve Balancer, EigenLayer Maker, Morpho, GMX + liquid staking tokens 11 adapters
solana/ Marinade, Jito, Orca Raydium, Drift, MarginFi Kamino, Jupiter Perps BlazeStake + 6 more 10 adapters
cost_basis_engine.py FIFO/LIFO/Average P&L Analytics
Support Services
ssl_service.py Certificate Mgmt
snapshot.py Portfolio Snapshots
logging_service.py Centralized Logs
taptools.py Cardano Token Data
transaction_history.py Multi-Chain Txns
defi.py DeFi Protocols
pricing.py Price Aggregation
moralis.py Multi-Chain Fallback
logostream.py Token Logo API
etherscan.py ETH Transaction History
log entries
SVG files
cert files
images
database.py SQLite Layer
Data Storage
static/demo-nfts/ Demo NFT SVGs
data/certs/ SSL Certificates (server.crt, server.key)
nft_images.db NFT Image Cache (thumbnails, metadata)
portfolio.db Main Database (users, sessions, wallets, balances, API keys, cache, transactions, nfts, security settings, logs)
Services Layer
Blockchain Services (34 Chains)
cardano.py Cardano (ADA)
ethereum.py Ethereum (ETH)
polygon.py Polygon (MATIC)
base.py Base (ETH)
evm_chain.py Generic EVM Service (BSC/Arbitrum/Avalanche Optimism/zkSync/Linea Scroll/Fantom/Cronos Gnosis/Moonbeam)
algorand.py Algorand (ALGO)
tron.py Tron (TRX)
vechain.py VeChain (VET)
cosmos.py Cosmos (ATOM)
near.py NEAR (NEAR)
icp.py ICP (ICP)
bitcoin.py Bitcoin (BTC)
solana.py Solana (SOL)
xrp.py XRP (XRP)
hedera.py Hedera (HBAR)
multiversx.py MultiversX (EGLD)
sui.py Sui (SUI)
aptos.py Aptos (APT)
filecoin.py Filecoin (FIL)
litecoin.py Litecoin (LTC)
dogecoin.py Dogecoin (DOGE)
zcash.py Zcash (ZEC)
tezos.py Tezos (XTZ)
stacks.py Stacks (STX)
NFT Services
nft_image_service.py Image Caching
nft.py Cardano NFTs
ethereum_nft.py Ethereum NFTs
solana_nft.py Solana NFTs
algorand_nft.py Algorand NFTs (ARC-3/69)
nft_scheduler.py Background Updates
HTTPS
HTTPS (fallback)
HTTPS (BSC/ARB/AVAX)
Cardano UTXOs
DeFi contract calls
Solana DeFi RPCs
Chainlink fallback
1st
2nd
3rd
4th
Security & Monitoring
.git/hooks/pre-push Git Integration
triggers
security_agent.py Pre-push Hook
runs
security_audit.py Automated Audit
scans
External APIs
Exchange APIs (42 Platforms)
Bitget Derivatives Exchange
Gate.io Comprehensive API
KuCoin Global Exchange
35 New Exchanges Bybit, MEXC, Kraken, Gemini Bitfinex, HTX, Crypto.com + 28 more
Coinbase CDP Portfolio Sync
Binance.com Global Exchange
Binance.US US Exchange
OKX Multi-Asset Exchange
Visual Assets
LogoKit Blockchain Logos
Logostream Token Logo API
Cardano Network
Blockfrost Blockchain Data (Primary)
CExplorer Staking Data
Maestro Blockchain Data (Fallback)
TapTools NFT Prices & Token Data
Koios Fallback API
EVM Chains (12 via Alchemy + Public RPCs)
Snowscan Avalanche Transactions
Etherscan Ethereum Transactions
BscScan BSC Transactions
Arbiscan Arbitrum Transactions
Polygonscan Polygon Transactions
Moralis Multi-Chain Fallback
Basescan Base Transactions
Alchemy ETH/Polygon/Base/BSC/ARB/AVAX +OP/zkSync/Linea/Scroll
Algorand Network
Tatum Algorand Data (Fallback)
Pera Wallet Algorand Data (Primary)
Tron Network
TronGrid TRX/TRC-20 Data (Free)
Other Blockchains
VeBlocks/Thor VeChain Data (Free)
Cosmos LCD PublicNode (Free)
NEAR RPC + NearBlocks (Free)
Rosetta API ICP Data (Free)
Mempool.space Bitcoin Fallback
Blockstream Bitcoin Data
Helius Solana RPC & NFTs
XRPL Cluster XRP Ledger API
Mirror Node Hedera API
MultiversX API MultiversX Data
Sui RPC Sui Data
Aptos API Aptos Data
Glif RPC Filecoin Data
BlockCypher LTC/DOGE Data (Free)
Blockchair Zcash Data (Free)
TzKT Tezos Data (Free)
Hiro API Stacks Data (Free)
Pricing (Fallback Chain)
4. DefiLlama Universal Fallback
1. CoinGecko Primary Price Data
3. Coinbase Spot Prices (Free)
2. CoinMarketCap Secondary Prices